

Pilot Results Draft — Qwen3-32B Value-Axis Transfer

(100-trajectory subsample; formatting reference)

Pilot note. These numbers are from a Stage-2 Colab pilot: all 14 successes in the ingested pool plus 86 subsampled failures (100 trajs total), frozen `value_axis_32b.npy`, primary layer 49, thinking ON. They are **not** the full verified-200 paper run. Branches below follow the Results scaffold outcome **C** (no frozen-axis transfer), with probes separating (decodability without axis alignment).

1 Results

1.1 Reconstruction and Data Overview

Our reconstruction of the value axis reproduces the published single-turn separation pattern on Qwen3-32B with thinking enabled: held-out AUROC of 0.872 at layer 49, with the strongest separation at layers 45–50, consistent with Jiang et al. at a lower absolute ceiling than their Qwen3-8B 0.95+ result. Layer 49 is fixed as the primary layer for all subsequent analyses.

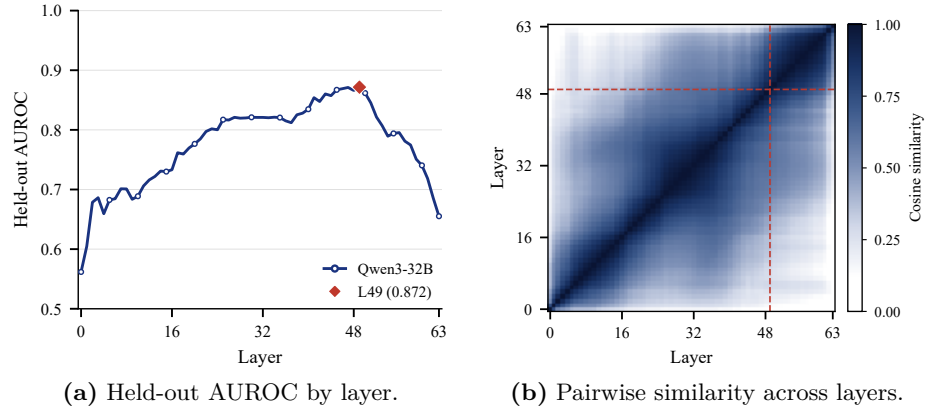


Figure 1: **The Qwen3-32B value axis generalizes to held-out ICRL criteria.** (a) Held-out AUROC peaks at layer 49 (0.872). (b) Axis directions are similar in middle-to-late layers; early and late directions are nearly orthogonal.

Pilot transfer sample. From 198 ingested verified trajectories (14 resolved / 184 unresolved), we keep **all 14 successes** and randomly subsample failures to a pilot of **100** trajectories (14 resolved, 86 unresolved; majority-class baseline 0.86). Exit statuses in the pilot: 97 **Submitted**, 3 **LimitsExceeded**. Trajectory lengths range from 4 to 59 steps (median 8, mean 12.3); resolved trajectories are shorter (median 7) than unresolved ones (median 8), and all trajectories with > 20 steps are unresolved ($n=17$). Because this is a one-seed design, no task has mixed outcomes across seeds (0 mixed-outcome tasks).

1.2 Final-Step Transfer

At the final step, the frozen axis **does not separate** outcomes: AUROC 0.488 [0.326, 0.651] against a majority-class baseline of 0.860, with the interval including chance (0.5). The task-level permutation test yields $p = 0.885$. The final-token robustness readout gives AUROC 0.492, consistent with the token-mean readout. Class means of `proj_mean` are essentially identical (unresolved 0.1345 vs. resolved 0.1334); the final-step histograms fully overlap (Figure 2).

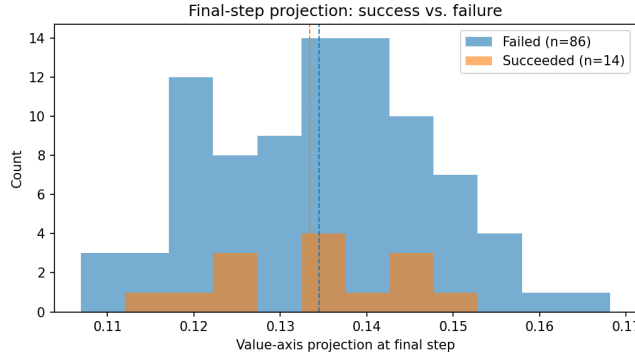


Figure 2: **Pilot: the frozen axis does not separate outcomes at the final step.** Layer-49 mean-over- G_t projections for resolved ($n=14$) vs. unresolved ($n=86$) trajectories.

1.3 Trajectory Evolution

Across relative-position bins, the frozen axis does **not** show the late-emerging transfer pattern hypothesized for a transferring value signal. Per-bin AUROCs (layer 49) are 0.339 [0.247, 0.442] (early), 0.265 [0.185, 0.385], 0.367 [0.182, 0.535], 0.457 [0.283, 0.622], and 0.492 [0.350, 0.644] (late). Early bins show *inverted* separation (AUROC $\ll$ 0.5, CIs entirely below chance); mid-to-late bins approach chance with intervals that include 0.5. Late-bin separation-to-noise is 0.023. Readout-noise check: `proj_mean` std 0.015 < `proj_final` std 0.025, as expected for averaging over G_t .

1.4 Layer Localization

Fixing the primary layer is not the sole explanation for the null. A final-step sweep over all 64 layers yields AUROCs mostly near chance in the “correct” direction; the maximum above 0.5 is only 0.615 (layer 44), and several late layers are strongly *anti*-aligned with success (e.g. layer 59 AUROC 0.292). Layer 49 (pre-registered from ICRL) sits at 0.488. We do **not** interpret inverted late-layer AUROCs as a transferring value signal without a pre-registered flip and held-out confirmation. Held-out trajectory re-selection for layer choice was not run in this pilot.

1.5 Internal Signal versus Verbalized Confidence

Not run in this pilot. Stage-4 post-hoc elicitation (verbalized $P(\text{success})$) was deferred. Under outcome C for the frozen axis, this comparison is secondary; we will report it for completeness on the full run if budget allows. *Scaffold branch C (placeholder)*: Neither signal separates outcomes until elicitation exists; the comparison is currently uninformative for the internal axis alone.

1.6 Fitted-Probe Reference

Fitted ℓ_2 logistic probes under task-level 5-fold CV achieve a grid-max AUROC of 0.935 at layer 51, relative-position bin 3 (0.7), and a **grid-mean** AUROC of 0.748 across all 64×5 cells (all cells used 5 folds). At the primary layer 49, bin-wise probe AUROCs are 0.769, 0.804, 0.725, 0.794, and 0.834 (late bin). The full layer-by-position grid is in the pilot CSV (`probe_auroc_grid.csv`).

Branch C / probes DO separate: Fitted probes separate outcomes where the frozen axis does not. Outcome information is present in agentic mean-pooled residuals but not along the original ICRL value axis — the representation, not the information, fails to transfer on this pilot. We treat `grid_max` as a summary only (selection over the layer $\times$ bin family can overfit); the elevated *mean* across the grid is the more informative decodability signal. Caveat: with only 14 successes, and with trajectory length correlated with failure, probe AUROCs may partly reflect length or other correlates; confound checks are required before a paper claim.

	Frozen axis (L49)	Fitted probes
Final / late readout AUROC	0.488 [0.326, 0.651]	0.834 (L49, late bin; CV mean)
Best cell (summary only)	—	0.935 (L51, bin 3)
Grid mean (all layers $\times$ bins)	—	0.748
Permutation p (final-step axis)	0.885	—

Table 1: **Pilot four-way summary:** frozen axis null, fitted probes decodable (outcome information present; not aligned with the ICRL axis).

1.7 Within-Task Contrast

Restricting to mixed-outcome tasks is **not applicable** in this one-seed pilot: 0 tasks have both resolved and unresolved rollouts. The within-task gap is undefined (`within_task_n_mixed = 0`). This contrast requires multi-seed data or a different design; we omit it rather than over-interpret the pooled null.